

Linear Algebra – MAT 2610

Section 4.3 (Linearly Independent Sets and Bases)

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Linear Independent/Dependent

An indexed set of vectors $\{v_1, \dots, v_n\}$ in V is said to be linearly independent if the vector equation

$$\underline{c_1 v_1 + \dots + c_n v_n = 0}$$

has *only* the trivial solution $c_1 = \dots = c_n = 0$

The set $\{v_1, \dots, v_n\}$ is said to be linearly dependent if $c_1 v_1 + \dots + c_n v_n = 0$ has a nontrivial solution, *i.e.*, if there are some values, $c_1, \dots, c_n$, *not all zero*, such that the equation holds



Theorem 4

An indexed set $\{v_1, \dots, v_p\}$ of two or more vectors, with $v_1 \neq 0$ is linearly dependent if and only if some v_j (with $j > 1$) is a linear combination of the preceding vectors $v_1, \dots, v_{j-1}$



Basis

Let H be a subspace of a vector space V . An indexed set of vectors $\mathcal{B} = \langle b_1, \dots, b_p \rangle$ in V is a **basis** for H if

- i. The set $\mathcal{B}$ is linearly independent; and
- ii. The subspace spanned by $\mathcal{B}$ is all of H , i.e.: $H = \text{Span}\{b_1, \dots, b_p\}$



Standard Basis

Let $\{e_1, \dots, e_n\}$ be the columns of the $n \times n$ matrix I_n . This is called the **standard basis** for $\mathbb{R}^n$.

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad \dots \quad e_p = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \leftarrow p^{\text{th}} \text{ row} \quad e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$



Theorem 5 – Spanning Set Theorem

Let $S = \{v_1, \dots, v_p\}$ be a set in V and let $H = \text{Span}(v_1, \dots, v_p)$

- a) If one of the vectors in S is a linear combination of the remaining vectors in S then the set formed from S by removing that vector still spans H
- b) If $H \neq \{0\}$ then some subset of S is a basis for H



Theorem 5 – Proof

Part a:

Rearrange the vectors in S if needed so that v_p is a linear combination of $\{v_1, \dots, v_{p-1}\}$ with $v_p = a_1 v_1 + \dots + a_{p-1} v_{p-1}$

Let $x = c_1 v_1 + \dots + c_p v_p$. Substitute the equation for v_p into the equation for x and you will get an expression for x as a linear combination of $\{v_1, \dots, v_{p-1}\}$

$$\begin{aligned} x &= c_1 v_1 + \dots + c_p v_p \\ &= c_1 v_1 + \dots + c_{p-1} v_{p-1} + c_p (a_1 v_1 + \dots + a_{p-1} v_{p-1}) \end{aligned}$$



Theorem 5 – Proof

Part b:

Assume S is linearly dependent (otherwise we are already done). Then one of the vectors of S can be removed ~~as~~ in part a. Repeat this procedure until the set of vectors is linearly independent. This is now a set spans H and is linearly independent, and thus is a basis.



Example

Let $A = \left\{ \underline{v_1} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \underline{v_3} = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix} \right\}$ be a generating set for the space H . These vectors are not linearly independent (why?)

Since $\underline{v_3 = 2v_1}$, Theorem 5 says that $\underline{\text{Span}\{v_1, v_2\}} = \text{Span}\{v_1, v_2, v_3\}$ and since $\{v_1, v_2\}$ is a linearly independent set, it is also a basis for H

$\left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \right\}$ is a basis for H



Column Space of a Matrix

Example: Find a basis for $\text{col}(A)$ where $A = \begin{bmatrix} 3 & 6 & 0 \\ 2 & 4 & 0 \\ 1 & 3 & 1 \end{bmatrix}$. $H = \text{Span} \left\{ \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 6 \\ 4 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

$$\begin{bmatrix} 3 & 6 & 0 \\ 2 & 4 & 0 \\ 1 & 3 & 1 \end{bmatrix} \xrightarrow{2r_1 - 3r_2 \rightarrow r_1} \begin{bmatrix} 0 & 0 & 0 \\ 2 & 4 & 0 \\ 1 & 3 & 1 \end{bmatrix} \xrightarrow{r_1 \leftrightarrow r_3} \begin{bmatrix} 1 & 3 & 1 \\ 2 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

A row of 0s means the vectors are linearly dependent. Need to remove a vector lin comb of others

$$\begin{bmatrix} 1 & 3 & 1 \\ 2 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{2r_1 - r_2 \rightarrow r_2} \begin{bmatrix} 1 & 3 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{2r_1 - 3r_2 \rightarrow r_1} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

This shows v_3 is a lin comb of v_1, v_2 , so basis = $\left\{ \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 6 \\ 4 \\ 3 \end{bmatrix} \right\}$



Theorem 6

The pivot columns of a matrix A form a basis for $Col(A)$



Theorem 6 - Proof

Let B be the reduced echelon form of A . The set of pivot columns of B is linearly independent (no pivot column can be a linear combination of the vectors before it).

Since A is row equivalent to B , the pivot columns of A are linearly independent as well (why?)

$$A \rightarrow B$$



Theorem 6 - Proof

Similarly, the non-pivot columns of A are linear combinations of the pivot columns (why?)

Then, the non-pivot columns can be discarded by Theorem 5, leaving the pivot columns as a basis for $Col(A)$



Row Space

Definition: The row space of a matrix A is the span of its rows, denoted by $\text{row}(A)$

Example: If $A \equiv \begin{bmatrix} 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 4 \end{bmatrix}$, then $\text{row}(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 2 \\ 4 \end{bmatrix} \right\}$



Theorem 7

If two matrices A and B are row equivalent, then their row spaces are the same. If B is in echelon form, the nonzero rows of B form a basis for the row space of A as well as for that of B .



Example

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 & 1 \\ 1 & 2 & 3 & 1 & 0 \\ -1 & -1 & 1 & 1 & -2 \\ 0 & 1 & 1 & 1 & -1 \end{bmatrix} \text{ has reduced form } \begin{bmatrix} 1 & 0 & 0 & -\frac{4}{3} & \frac{7}{3} \\ 0 & 1 & 0 & \frac{2}{3} & -\frac{2}{3} \\ 0 & 0 & 1 & \frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Find a basis for the row space of A , the column space of A and the null space of A

$$\text{Basis row}(A) = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ -4/3 \\ 7/3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 2/3 \\ -2/3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1/3 \\ -1/3 \end{bmatrix} \right\}$$

$$\text{Basis col}(A) = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ 1 \\ 1 \end{bmatrix} \right\}$$



$$\begin{array}{ccccc} x_1 & x_2 & x_3 & x_4 & x_5 \\ \left[\begin{array}{ccccc} 1 & 0 & 0 & -\frac{4}{3} & \frac{7}{3} \\ 0 & 1 & 0 & \frac{2}{3} & -\frac{2}{3} \\ 0 & 0 & 1 & \frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \end{array}$$

$$x_1 - \frac{4}{3}x_4 + \frac{7}{3}x_5 = 0$$

$$x_2 + \frac{2}{3}x_4 - \frac{2}{3}x_5 = 0$$

$$x_3 + \frac{1}{3}x_4 - \frac{1}{3}x_5 = 0$$

 $\Rightarrow$

$$x_1 = \frac{4}{3}x_4 - \frac{7}{3}x_5$$

$$x_2 = -\frac{2}{3}x_4 + \frac{2}{3}x_5$$

$$x_3 = -\frac{1}{3}x_4 + \frac{1}{3}x_5$$

$$x_4 = x_4$$

$$x_5 = x_5$$

$$\text{Nul}(A) = x_4 \begin{bmatrix} 4/3 \\ -2/3 \\ -1/3 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -7/3 \\ 2/3 \\ 1/3 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{Basis}(\text{Nul}(A)) = \left\{ \begin{bmatrix} 4/3 \\ -2/3 \\ -1/3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -7/3 \\ 2/3 \\ 1/3 \\ 0 \\ 1 \end{bmatrix} \right\}$$



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